SMOOTHNESS OF A RELATIVE GLOBAL COMPLETE INTERSECTION IS DETECTED BY A NONVANISHING MAXIMAL MINOR

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The shape of the naive cotangent complex of a relative global complete intersection. Invoked in the counted proof at lines 37860-37867. Identifies the complex with an explicit two-term complex given by the Jacobian matrix.
- Localising the naive cotangent complex. Invoked twice in the counted proof (lines 37876 and 37885). Relates smoothness after inverting an element to the localized complex being split.

Source reference. Original lemma 00S7: sources/stacks-algebra/pinned-original-source.tex, source lines 35909–35909. Navigation pointer for \ref{lemma-localize-NL} used inside the counted span (line 37876); the argument that consumes it is retained in the span.

Source reference. Original lemma 00SV: sources/stacks-algebra/pinned-original-source.tex, source lines 36984–36984. Navigation pointer for \ref{lemma-relative-global-complete-intersection-conormal} used inside the counted span (line 37860); the argument that consumes it is retained in the span.

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let R be a ring. Let $S = R[x_1, \ldots, x_n]/(f_1, \ldots, f_c)$ be a relative global complete intersection. Let $\mathfrak{q} \subset S$ be a prime. Then $R \to S$ is smooth at $\mathfrak{q}$ if and only if there exists a subset $I \subset \{1, \ldots, n\}$ of cardinality c such that the polynomial*

$$g_I = \det(\partial f_j/\partial x_i)_{j=1,\ldots,c,\ i \in I}.$$

does not map to an element of $\mathfrak{q}$.

Proof. By Lemma 00SV we see that the naive cotangent complex associated to the given presentation of S is the complex

$$\bigoplus\nolimits_{j=1}^{c} S \cdot f_j \longrightarrow \bigoplus\nolimits_{i=1}^{n} S \cdot \mathrm{d}x_i, \quad f_j \longmapsto \sum \frac{\partial f_j}{\partial x_i} \mathrm{d}x_i.$$

The maximal minors of the matrix giving the map are exactly the polynomials g_I.

Assume g_I maps to $g \in S$, with $g \notin \mathfrak{q}$. Then the algebra S_g is smooth over R. Namely, its naive cotangent complex is quasi-isomorphic to the complex above localized at g, see Lemma 00S7. And by construction it is quasi-isomorphic to a free rank $n - c$ module in degree 0.

Conversely, suppose that all g_I end up in $\mathfrak{q}$. In this case the complex above tensored with $\kappa(\mathfrak{q})$ does not have maximal rank, and hence there is no localization by an element $g \in S$, $g \notin \mathfrak{q}$ where this map becomes a split injection. By Lemma 00S7 again there is no such localization which is smooth over R. □